

Engineering Physics (2025)

Course code 25PY101

M2T1

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M2U1 Plan

Quantum Free Electron Theory

Fermi-Dirac distribution

Electronic specific heat of solids

Density of states (qualitative)

Success and Failures of quantum free electron theory of solids

E-k diagram

Quantum Band Theory: Classification of materials based on bands in solids

Fermi level in semiconductors- intrinsic and extrinsic

p-n junction diode – forward and reverse conditions

Solar cell

Photo-diode

Direct and indirect band gap semiconductors

Light Emitting Diode

Laser

Optical fiber

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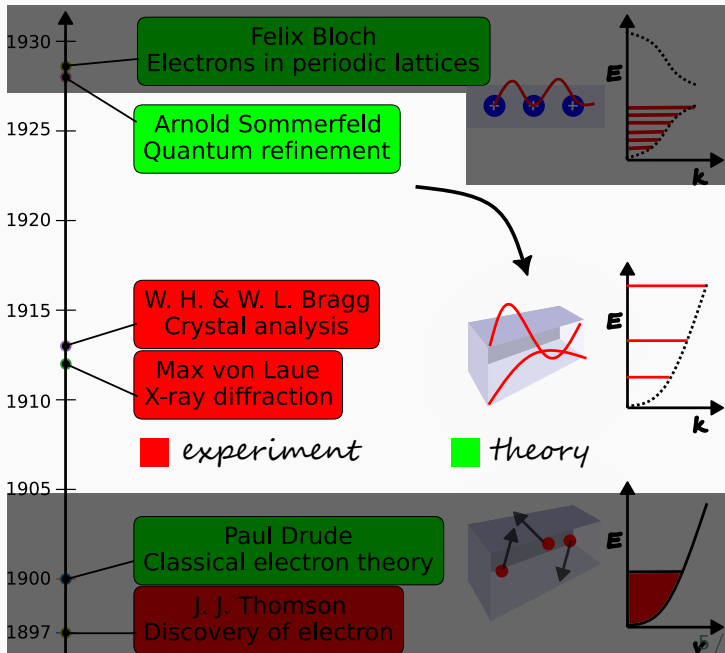
Optical fiber

Summary of Classical free electron theory (CFET)

- Derived $\sigma = \frac{ne^2\tau}{m}$.
- Connected microscopic electron properties to macroscopic Ohm's law.
- Classical free electron theory has qualitative and quantitative agreement with experiment values of conductivity.
- However, the theory has drawbacks. Some of them are
 - Incorrect prediction for conductivity vs valency.
 - Cannot explain anomalous sign of Hall coefficient in some metals.
 - Underestimation of mean free path.
 - Cannot explain classification of materials into conductors. semi-conductors and insulators.
 - Wrong prediction of conductivity vs temperature.
 - Overestimation of heat capacity.

Electron theories of metals

1. Classical free electron theory
2. Quantum free electron theory
3. Quantum band theory



Pauli's exclusion principle

Postulate

- No two electrons can occupy the same quantum state.
- No two electrons can share the same set of quantum numbers.
- Electron obeys Pauli's exclusion principle since it has spin $\frac{1}{2}$.



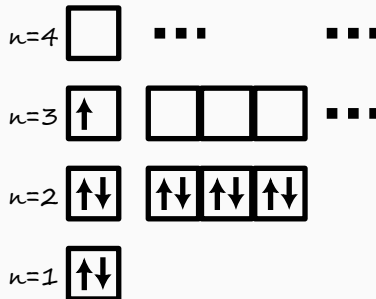
W. Pauli

Definition

A quantum state is defined by a set of quantum numbers.

Problem

What quantum state is occupied by valence electron in Na?



Postulates

1. **Waves:** Electrons are quantum waves with wavevector k , angular frequency ω .
2. **Fermion:** Electron is a spin $\frac{1}{2}$ particle and obeys Pauli's exclusion principle.
3. **Independent electron approximation:** Electrons are independent and mutual repulsion between them is ignored.
4. **Free electron approximation:** Electrons are free and move in an infinite potential well.
5. **Quantum Thermodynamics:** The thermalization is governed by Fermi-Dirac statistics.

Density of states $Z(E) = \frac{d(\frac{N}{V})}{dE}$

Definition

- Density of states $Z(E)$ ¹ is defined as the rate of change of the number of states per unit volume upto energy E with respect to energy E .
- Therefore, the the number of states per unit volume from energy E to $E + dE$ is given by

$$Z(E) dE.$$

Number of states per unit volume is

$$\rho_N(E) = \frac{N}{V} = \frac{\pi}{3} \left(\frac{\sqrt{8m}}{h} \right)^3 E^{3/2}$$

¹In Neamen, density of states is denoted by $g(E)$.

DOS in textbook form

Differentiate $\rho_N(E)$ with respect to E to get the density of states function:

$$Z(E) = \frac{d\rho_N(E)}{dE} = \frac{\pi}{2} \cdot \left(\frac{\sqrt{8m}}{h} \right)^3 E^{1/2}$$

$$Z(E) = \frac{\pi}{2} \cdot 8 \cdot \left(\frac{2m}{h^2} \right)^{3/2} E^{1/2}$$

$$\therefore Z(E) = 4\pi \cdot \left(\frac{2m}{h^2} \right)^{3/2} E^{1/2}.$$

Key Insight

For 3D infinite potential, $Z(E) \propto \sqrt{E}$.

Problem

Find the form of $Z(E)$ for 2D infinite potential.



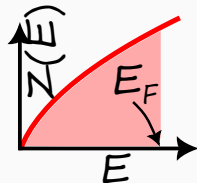
Fermi level

Definition

Fermi level of a metal is the maximum energy that an electron can have at $T = 0$ K.

At absolute zero all states are filled up to the Fermi energy E_F . The total electron density n_c (electrons per unit volume) is

$$\begin{aligned}n_c &= \frac{N}{V} = \int_0^{E_F} Z(E) dE = 4\pi \cdot \left(\frac{2m}{h^2}\right)^{3/2} \int_0^{E_F} E^{1/2} dE \\&= 4\pi \cdot \left(\frac{2m}{h^2}\right)^{3/2} \cdot \frac{2}{3} E_F^{3/2} \\&= \frac{8\pi}{3} \left(\frac{2m}{h^2}\right)^{3/2} E_F^{3/2}.\end{aligned}$$



Relating Fermi energy to electron density

Invert the relation to get the Fermi energy as a function of density:

$$E_F^{3/2} = \left(\frac{h^2}{2m} \right)^{3/2} \cdot \frac{3n_c}{8\pi}.$$

$$\boxed{\therefore E_F = \frac{h^2}{2m} \left(\frac{3n_c}{8\pi} \right)^{2/3} .}$$

This is the Fermi energy of a free electron gas in three dimensions at $T = 0$.

Problem

Calculate E_F for metal with electron density $n_c = 5.8 \times 10^{28} \text{ m}^{-3}$.

Test Your Understanding

1. What are the postulates of QFET.
2. Define Pauli's exclusion principle.
3. Define Fermi energy level.
4. Define density of states function.

- Pauli's exclusion principle
- Fermi level
- Density of states
- Fermi velocity
- Quantum state
- energy level
- degeneracy
- multiplicity

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Quantum thermodynamics: Fermi–Dirac (FD) Statistics

- Describes occupancy of energy states by electrons that obey **Pauli exclusion principle**.
- The probability of occupation of an energy level E at temperature T and Fermi energy level E_F is given by

$$P_{FD}(E) = \frac{1}{\exp\left(\frac{E-E_F}{k_B T}\right) + 1}.$$

Here k_B is the Boltzmann constant equal to $1.38 \times 10^{-23} \text{ J K}^{-1}$.

- To understand the FD statistics, let us analyze the probability function in two cases –
 1. $T \rightarrow 0$ limit
 2. $T \neq 0$

Key Insight

The Fermi-Dirac function is a result of Pauli exclusion principle.



Fermi–Dirac Statistics at the $T \rightarrow 0$ limit

- Take limit $T \rightarrow 0$ in definition:

$$P_{FD}(E) = \lim_{T \rightarrow 0} \frac{1}{\exp\left(\frac{E-E_F}{k_B T}\right) + 1} = \frac{1}{\exp\left(\frac{E-E_F}{0}\right) + 1}.$$

- This limit needs to be analyzed for three cases –
 1. $E < E_F$
 2. $E > E_F$
 3. $E = E_F$

Fermi–Dirac Statistics at the $T \rightarrow 0$ limit

1. $E < E_F$

$$\begin{aligned}P_{FD}(E) &= \frac{1}{\exp\left(\frac{-ve}{0}\right) + 1} \\&= \frac{1}{\exp(-\infty) + 1} \\&= \frac{1}{0 + 1} \\&= \frac{1}{1} \\&= 1\end{aligned}$$

2. $E > E_F$

$$\begin{aligned}P_{FD}(E) &= \frac{1}{\exp\left(\frac{+ve}{0}\right) + 1} \\&= \frac{1}{\exp(\infty) + 1} \\&= \frac{1}{\infty + 1} \\&= \frac{1}{\infty} \\&= 0\end{aligned}$$

3. $E = E_F$

$$P_{FD}(E) = \frac{1}{\exp\left(\frac{0}{0}\right) + 1}$$

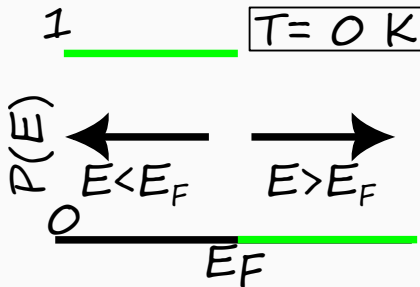
- Since $\frac{0}{0}$ is indeterminate form, $P_{FD}(E)$ is not defined at $T = 0$ K, $E = E_F$.

Fermi–Dirac Statistics at the $T \rightarrow 0$ limit

- Hence, at $T = 0$ K the distribution becomes a step function

$$P_{FD}(E) = \begin{cases} 1, & E < E_F \\ 0, & E > E_F \end{cases}$$

where E_F is the Fermi energy.



Key Insight

At $T = 0$ K, the Fermi-Dirac distribution is a step function.



Fermi–Dirac Statistics at $T \neq 0$

- At $T \neq 0$, consider $k_B T$ is a small positive quantity +ve.

$$P_{FD}(E) = \frac{1}{\exp\left(\frac{E-E_F}{+ve}\right) + 1}.$$

- This limit needs to be analyzed for three cases ²
 1. $E \ll E_F$ so that $E - E_F$ is a big negative number i.e. --ve
 2. $E \gg E_F$ so that $E - E_F$ is a big positive number i.e. ++ve
 3. $E = E_F$

² $\gg$ is symbol for very greater than and $\ll$ is symbol for very lesser than

Fermi–Dirac Statistics at $T \neq 0$

1. $E \ll E_F$ ^a

2. $E \gg E_F$

3. $E = E_F$

$$\begin{aligned} P_{FD}(E) &= \frac{1}{\exp\left(\frac{-ve}{+ve}\right) + 1} \\ &\simeq \frac{1}{\exp(-\infty) + 1} \\ &= \frac{1}{0 + 1} \\ &= \frac{1}{1} \\ &= 1 \end{aligned}$$

$$\therefore P_{FD}(E) \simeq 1$$

$$\begin{aligned} P_{FD}(E) &= \frac{1}{\exp\left(\frac{++ve}{+ve}\right) + 1} \\ &\simeq \frac{1}{\exp(\infty) + 1} \\ &= \frac{1}{\infty + 1} \\ &= \frac{1}{\infty} \\ &= 0 \end{aligned}$$

$$\therefore P_{FD}(E) \simeq 0$$

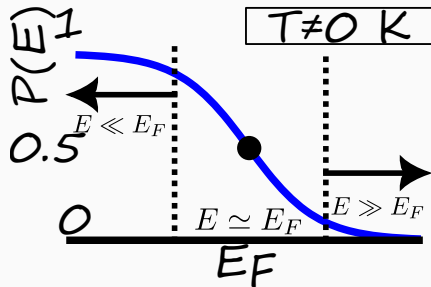
$$\begin{aligned} P_{FD}(E) &= \frac{1}{\exp\left(\frac{0}{+ve}\right) + 1} \\ &= \frac{1}{1 + 1} \\ &= \frac{1}{2} \end{aligned}$$

^a $\simeq$ is symbol for nearly equal.

Fermi–Dirac Statistics at $T \neq 0$

- Hence, at $T \neq 0$ K the distribution becomes a “smeared” ³ step function

$$P_{FD}(E) = \begin{cases} \simeq 1 & E \ll E_F \\ \simeq 0 & E \gg E_F \end{cases}$$



Key Insight

At $T = 0$ K, the Fermi-Dirac distribution is a “smeared” step function.

³smeared means smoothened.



Problems on Fermi Dirac statistics

Problem

Calculate the probability that an energy level $3k_B T$ above the Fermi energy is occupied by an electron

Problem

Calculate the probability that an energy level $3k_B T$ below the Fermi level is empty

Problem

Fermi energy of a metal is 6.25 eV. Calculate the temperature at which there is a 1 % probability that a state 0.30 eV below the Fermi energy level will not contain an electron.

Problem

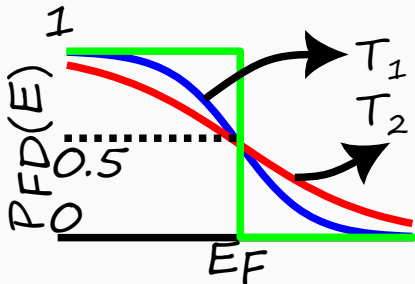
Calculate the temperature at which the probability that an energy level 0.3 eV below the Fermi level is occupied is 0.7.

Problems on Fermi Dirac statistics

Problem

Calculate the temperature at which the probability that an energy level $3k_B T$ above the Fermi level is occupied is 0.7.
[Hint: This is a wrong question]

Problem



Statement: $T_1 > T_2$.

Is the statement true or false?

Test Your Understanding

1. Explain the variation of Fermi-Dirac distribution function at $T = 0\text{ K}$ and $T \neq 0\text{ K}$.
2. Show that the probability of finding an electron of energy ΔE above the Fermi level is same as the probability of not finding an electron at energy ΔE below the Fermi level.
3. Show that the occupancy probabilities of two states whose energies are equally spaced above and below the Fermi energy add up to one.

- occupied level
- unoccupied level
- probability of occupancy
- probability of vacancy

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Specific heat capacity

Definition

- Specific heat capacity ⁴ C is defined as the rate of change of energy E with temperature T per unit mole of material.

$$C = \frac{dE}{dT} \quad \text{SI unit} \quad [C] = \left[\frac{E}{T} \cdot \frac{1}{\text{mole}} \right] = \text{J K}^{-1} \text{mol}^{-1}.$$

- Metals heat “quickly”. When we heat a metal, the heat energy is transferred to electrons. The quickness is measured in terms of heat capacity.

Problem

Specific heat of mercury is $0.14 \text{ J g}^{-1} \text{ K}^{-1}$, water is $1 \text{ cal g}^{-1} \text{ }^{\circ}\text{C}^{-1}$, ethanol is $2.44 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$. Which has more specific heat? Which better material for thermometer?

[$^{200}_{80}\text{Hg}$, $1 \text{ cal} = 4.2 \text{ J}$, ethanol = $\text{CH}_3\text{CH}_2\text{OH}$]

⁴It is also called simply as specific heat.

Specific heat capacity – CFET

- Energy of electron at temperature T is $E_{\text{el}} = \frac{3}{2}k_B T$.
- Energy of a mole of electrons is

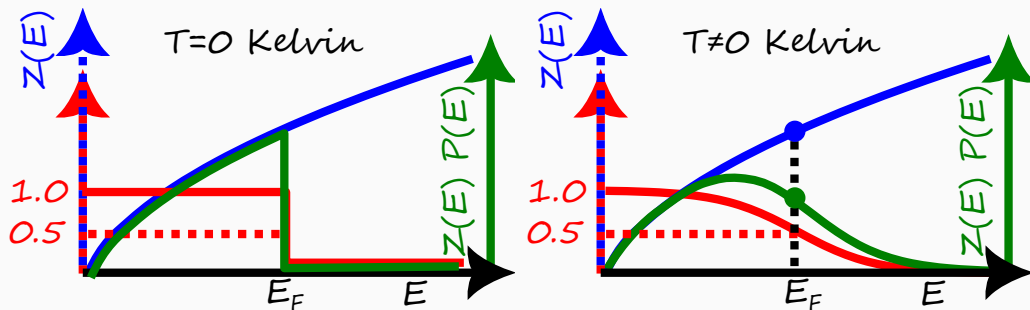
$$E = E_{\text{el}} N_A = \frac{3}{2}k_B N_A T = \frac{3}{2}RT. \quad [R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}]$$

- Therefore, specific heat capacity is

$$C_{\text{theory}}^{\text{classical}} = \frac{dE}{dT} = \frac{3}{2}R \sim 12 \text{ J mol}^{-1} \text{ K}^{-1}.$$

- But experimental values of specific heat capacity $C_{\text{experiment}}$ are **smaller by two orders** of magnitude.

Specific heat capacity – QFET



- For the derivation of specific heat capacity, only the **form** of density of states functions is necessary and not the exact expression. For 3D metal, $Z(E) \propto \sqrt{E}$ so that

$$Z(E) = \alpha \sqrt{E}, \quad \text{where } \alpha \text{ is a constant}$$

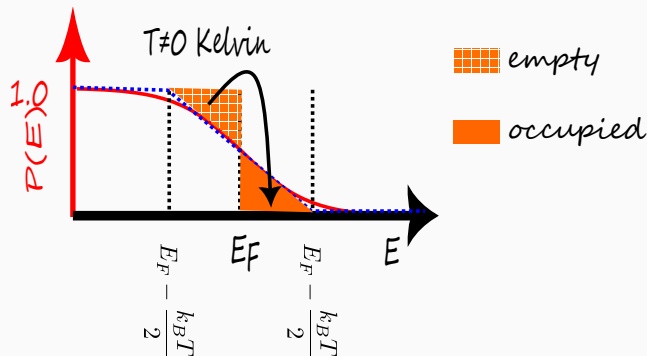
Specific heat capacity – Form of $Z(E)$

- At $T = 0$ K, electron has energy less than or equal to Fermi energy level.
- The number of electrons per unit volume n_c at $T = 0$ K is given by

$$\begin{aligned}n_c &= \int_0^{E_F} Z(E) P(E) dE \\&= \int_0^{E_F} Z(E) dE \\&= \int_0^{E_F} \alpha \sqrt{E} dE \\&= \frac{2}{3} \alpha E_F^{3/2}\end{aligned}$$

$$\therefore n_c = \frac{2}{3} \alpha E_F^{3/2}.$$

Specific heat capacity – Fraction of thermalized electrons



- The number of thermalized electrons is equal to area of the filled triangle multiplied by the density of states in the neighbourhood.
- Area of the filled triangle is

$$A = \frac{1}{2} \frac{k_B T}{2} \frac{1}{2} = \frac{1}{8} k_B T$$

$$\therefore n_{\text{thermal}} = \frac{1}{8} k_B T \cdot Z(E_F) = \frac{1}{8} \alpha k_B T \sqrt{E}$$

Specific heat capacity – Energy of thermalized electron

- The net increase in energy of thermalized electron is

$$\Delta E_{\text{thermalized}} = E_F + \frac{k_B T}{2} - \left(E_F - \frac{k_B T}{2} \right) = k_B T$$

- Since only a fraction of the electrons are thermalized, the net increase in energy of 1 electron is

$$\begin{aligned}\Delta E_{1 \text{ el}} &= \Delta E_{\text{thermalized}} \cdot f_{\text{thermalized}} \\ &= \frac{3}{16} k_B T \cdot \frac{k_B T}{E_F}\end{aligned}$$

- Therefore, the net increase in energy of 1 mole of electrons is

$$\begin{aligned}\Delta E_{1 \text{ mole}} &= \Delta E_{1 \text{ el}} \cdot N_A \\ &= \frac{3}{16} (k_B N_A) T \cdot \frac{k_B T}{E_F} \\ &= \frac{3}{16} R \frac{k_B T^2}{E_F}\end{aligned}$$

Specific heat capacity – Electronic contribution

- The specific heat capacity is given by

$$\begin{aligned}C_{\text{theory}}^{\text{quantum}} &= \frac{dE_{\text{el}}}{dT} \\&= \frac{3}{8} R \frac{k_B T}{E_F} \\&= \frac{3}{2} R \cdot \left(\frac{1}{4} \frac{k_B T}{E_F} \right) \\&= C_{\text{theory}}^{\text{classical}} \cdot \left(\frac{1}{4} \frac{k_B T}{E_F} \right)\end{aligned}$$

$$\therefore C_{\text{theory}}^{\text{quantum}} \simeq \left(\frac{k_B T}{E_F} \right) \cdot C_{\text{theory}}^{\text{classical}}.$$

- Therefore, quantum theory corrects the classical expression by the fraction $\frac{k_B T}{E_F}$. At room temperature, specific heat capacity is given by

$$C_{\text{theory}}^{\text{quantum}} = C_{\text{theory}}^{\text{classical}} \cdot \frac{k_B T_{300\text{K}}}{E_F} \simeq 0.01 C_{\text{theory}}^{\text{classical}} \simeq C_{\text{experiment}}$$

This validates Quantum Free Electron Theory.

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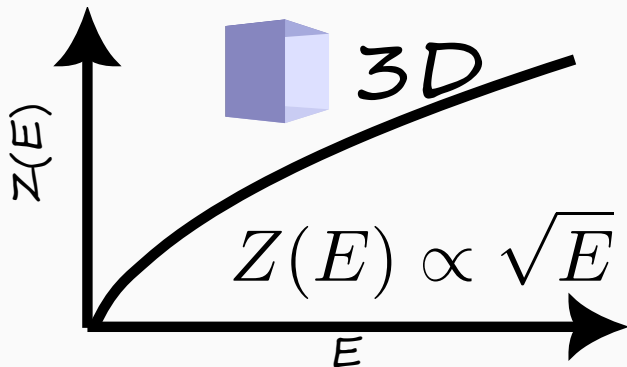
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Density of states (3D)

- The form of the DOS in 3D is $Z(E) \propto \sqrt{E}$



Problem

If $Z_{3D}(E) = k\sqrt{E}$, where k is a constant, find the expression for conduction electron number density n_c in terms of Fermi level E_F at $T = 0\text{ K}$ in a 3D material.

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Merits and demerits of QFET

Merits

- Electrical conductivity
- Thermal conductivity
- Heat capacity

Demerits

- Cannot explain classification of condensed matter into metals, semiconductors and insulators
- Occurrence of positive Hall coefficient in some metals like Zn.

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$E - k$ diagram – 1D case

- In a 1D metal, the electron is free as well as independent.
- For a free electron, the energy is related to the momentum p by

$$E = \frac{p^2}{2m}.$$

- However, since the electron behaves also as a wave, its wavelength λ is related to the momentum by the de Broglie relation

$$\lambda = \frac{h}{p} \quad \Rightarrow \quad p = \frac{h}{\lambda}.$$

- By definition, the wavevector k of the electron is inversely related to the wavelength as

$$k = \frac{2\pi}{\lambda}$$

- Therefore, the energy is related to the wavevector by

$$E = \frac{1}{2m} \left(\frac{h}{\lambda} \right)^2 = \frac{1}{2m} \left(\frac{hk}{2\pi} \right)^2 = \frac{\hbar^2 k^2}{2m}.$$

Summary of Quantum free electron theory (QFET)

- Derived density of states function $Z(E)$.
- Quantum free electron theory addressed the electronic contribution to specific heat of metals
- However, the theory has drawbacks. Some of them are
 - Cannot explain anomalous sign of Hall coefficient in some metals.
 - Cannot explain classification of materials into conductors, semi-conductors and insulators.

Reason for drawbacks

- Invalidity of free electron approximation.
- Drawbacks addressed by Bloch's quantum band theory.

Test Your Understanding

1. If the Fermi level of metal is E_f , what is the fraction of thermalized electrons at temperature T .
2. A thermalized electron according to CFET increases the energy of metal by $\mathcal{O}(k_B T)$. However, according to QFET, the thermalized electron increases the energy of metal by $\mathcal{O}(fk_B T)$, where f is the fraction of thermalized electrons. Explain.
3. Prove that the dispersion relation $E - k$ in metals is given by

$$E(k) = \frac{\hbar^2 k^2}{2m}.$$

4. List the merits and demerits of QFET.

- specific heat capacity C
- ideal gas constant R
- thermalized electron
- $E - k$ dispersion relation

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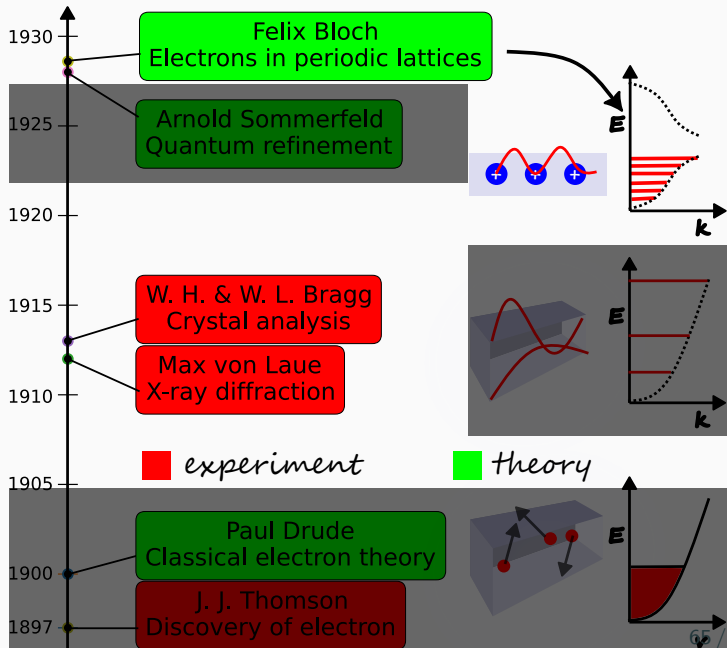
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2. Quantum free electron theory

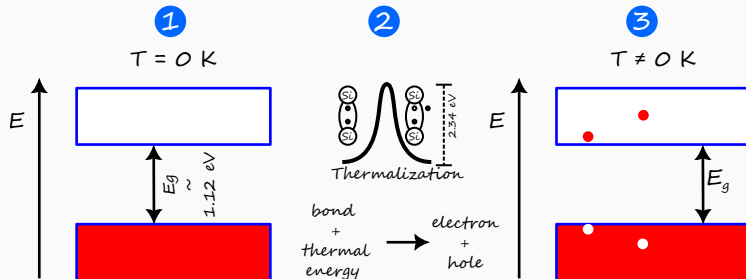
3. Quantum band theory (QBT)



Postulates

1. **Waves:** Electrons are quantum waves with wavevector k , angular frequency ω .
2. **Fermion:** Electron is a spin $\frac{1}{2}$ particle and obeys Pauli's exclusion principle.
3. **Independent electron approximation:** Electrons are independent and mutual repulsion between them is ignored.
4. **Periodic potential approximation:** Electrons move in a periodic potential.
5. **Quantum Thermodynamics:** The thermalization is governed by Fermi-Dirac statistics.

Energy band diagram



1. At $T = 0\text{ K}$,
 - The topmost occupied band is called **valence band**.
 - The lower most unoccupied is called **conduction band**.
 - These two bands are separated by a forbidden gap called the **energy band gap**.
2. When temperature is increased, the covalent bond is broken due to thermalization. This results in the formation of **electron-hole pair** ⁵.
3. At $T \neq 0\text{ K}$, the electron occupies energy level in the conduction band whereas the hole occupies energy level in the valence band.

⁵Hole is the absence of electron in the valence band.

Classification of solids

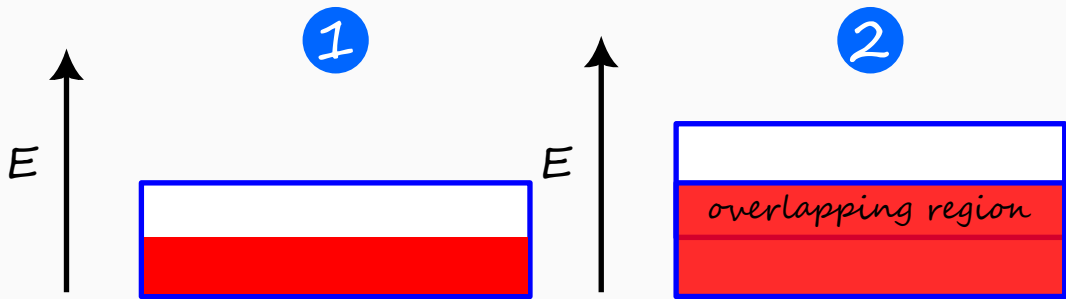
- According to band theory, conductivity of solid is determined by the band gap separating the conduction band and valence band.
- Based on the band gap, the solids are classified into
 1. metals or conductors
 2. semiconductors
 3. insulators

Key Insight

The classification of solids is based on energy band gap.

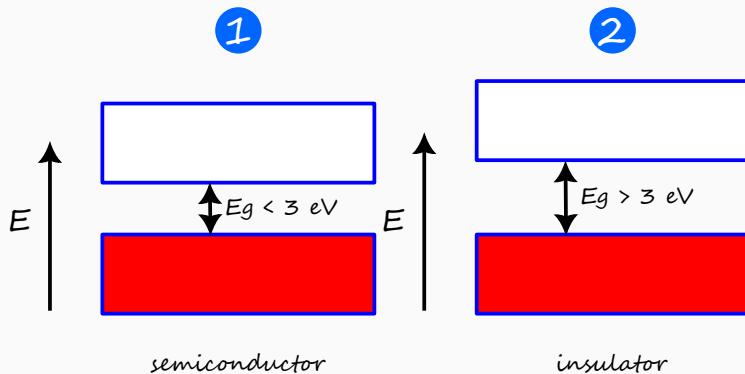


Energy band diagram – Conductor



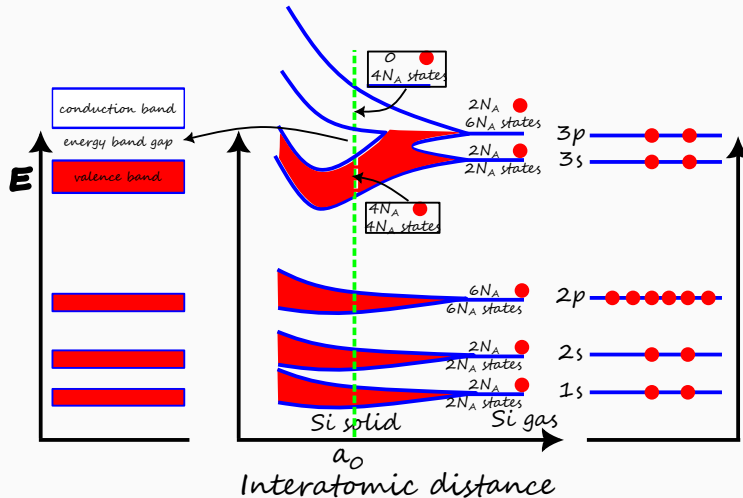
- In conductors, there are two cases
 1. the conduction band is half filled.
 2. the conduction band overlaps with empty upper band.
- In both cases, there is no band gap so that empty states are available for conduction.
- Since there is no band gap, there is no absence of electron at $T \neq 0$. Therefore, there are no holes in conductors.

Energy band diagram – Semiconductor



- In either semiconductor or insulator, there is a band gap so that empty states are not available for conduction.
 1. The energy band gap is less than 3 eV for semiconductor.
 2. The energy band gap is greater than 3 eV for insulator

Energy band diagram – Si



Key Insight

The valence band is completely filled and the conduction band is completely empty at $T = 0\text{ K}$.



Test Your Understanding

1. What are the postulates of QBT?
2. Define the periodic potential approximation.
3. In the parabolic approximation, since the conduction band is approximated by an upward parabola, the conduction band charge carrier is electron-like. Explain.
4. Hole has a negative effective mass. Explain.

Technical terms

- periodic potential approximation
- forbidden energy level
- band structure
- conduction band
- valence band
- energy band gap
- energy band diagram
- conduction band minimum E_c
- valence band maximum E_v
- parabolic approximation
- effective mass of electron and hole

M2U1 Plan

Quantum Free Electron Theory

Fermi-Dirac distribution

Electronic specific heat of solids

Density of states (qualitative)

Success and Failures of quantum free electron theory of solids

E-k diagram

Quantum Band Theory: Classification of materials based on bands in solids

Fermi level in semiconductors- intrinsic and extrinsic

p-n junction diode – forward and reverse conditions

Solar cell

Photo-diode

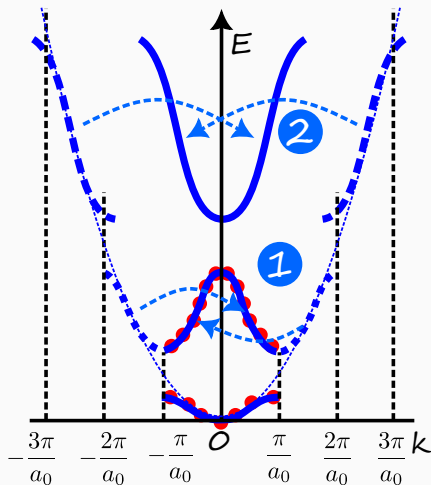
Direct and indirect band gap semiconductors

Light Emitting Diode

Laser

Optical fiber

Reduced band dispersion – Band structure



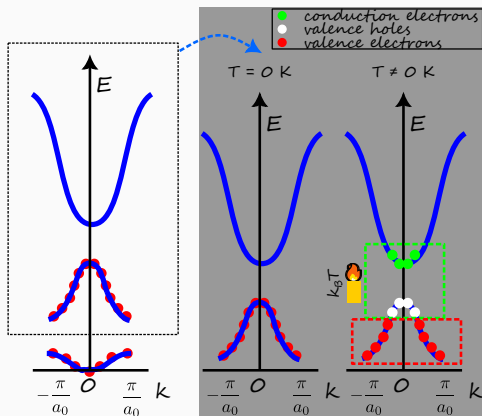
- Due to the periodicity of lattice, the band dispersion can be analyzed within the **zone** from $k = -\frac{\pi}{a_0}$ to $k = +\frac{\pi}{a_0}$.
- The original band dispersion can be reduced to the zone by folding the “arms” of the parabola resulting in the reduced band dispersion. This process is called **zone folding**.
- The reduced band dispersion is called the **band structure** of the material and it is the signature of the material.

Key Insight

Bandstructure is the fingerprint or “DNA” of a crystalline solid.



Band structure – Concept of hole



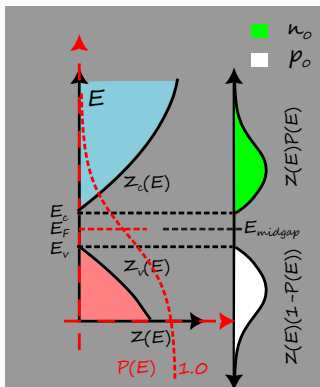
- For a semiconductor, due to the presence of energy band gap, there are no free electrons at $T = 0 \text{ K}$.
- At $T \neq 0 \text{ K}$, the electrons near the valence band maximum thermalize and jump to the levels near the conduction band minimum. This results in quantum vacancy in the valence band. This is called **hole**.
- Conduction electron and valence hole contribute to conduction.

Key Insight

- Hole is the absence of electron in the valence band.
- Hole is **not a physical reality**.



Electron carrier concentration



- The conduction electron density n_0 is given by

$$n_0 = \int_{E_c}^{\infty} Z_c(E) P(E) dE$$

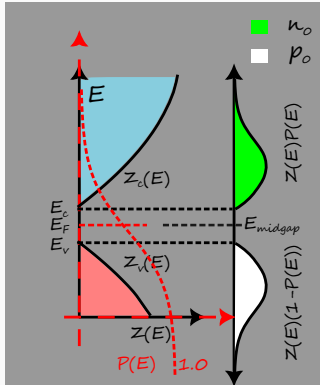
$$= \int_{E_c}^{\infty} \frac{\alpha^* \sqrt{E - E_c}}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)} dE$$

- By applying Boltzmann approximation and the exact expression for α^* , it can be shown that

$$n_0 = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)$$

where N_c is the **effective density of states** in the conduction band.

Hole carrier concentration



- Similarly the valence hole density p_0 is given by

$$p_0 = \int_{E_c}^{\infty} Z_c(E) (1 - P(E)) dE$$

$$= \int_0^{E_v} \beta^* \sqrt{E_v - E} \left(1 - \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)} \right) dE$$

- By applying **Boltzmann approximation** and the exact expression for β^* , it can be shown that

$$p_0 = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)$$

where N_v is the **effective density of states** in the valence band.

Intrinsic carrier concentration – $n_0 p_0$ product

- For an intrinsic semiconductor, electron-hole pairs are generated at $T \neq 0$ K. So the conduction electron density is equal to valence hole density

$$n_0 = p_0 = n_i$$

where n_i is defined as the intrinsic carrier concentration.

- The intrinsic carrier concentration is given by the product of electron and hole concentrations

$$\begin{aligned} n_0 p_0 = n_i^2 &= N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) \cdot N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right) \\ &= N_c N_v \exp\left(-\frac{E_c - E_F + E_F - E_v}{k_B T}\right) \\ n_i^2 &= N_c N_v \exp\left(-\frac{E_c - E_v}{k_B T}\right) \end{aligned}$$

Intrinsic carrier concentration

- Therefore, the $n_0 p_0$ is independent of Fermi energy level and is given in terms of the material properties and temperature.

$$n_0 p_0 = n_i^2$$

where n_i is given by

$$\begin{aligned} n_i &= \sqrt{N_c N_v \exp\left(-\frac{E_c - E_v}{k_B T}\right)} \\ &= \sqrt{N_c N_v} \exp\left(-\frac{E_c - E_v}{2k_B T}\right) \\ &= \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right) \end{aligned}$$

where E_g is the energy band gap and is given by

$$E_g = E_c - E_v$$

$$\therefore n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right)$$

Fermi level of intrinsic semiconductor

- Since the electron and hole exist in pairs in an intrinsic semiconductor

$$\begin{aligned} n_0 &= p_0 \\ \Rightarrow N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) &= N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right) \end{aligned}$$

- Taking logarithm on both sides of the equation, we have

$$\begin{aligned} \ln N_c + \ln \left[\exp\left(-\frac{E_c - E_F}{k_B T}\right) \right] &= \ln N_v + \ln \left[\exp\left(-\frac{E_F - E_v}{k_B T}\right) \right] \\ \Rightarrow \ln N_c - \frac{E_c - E_F}{k_B T} &= \ln N_v - \frac{E_F - E_v}{k_B T} \\ \Rightarrow \frac{2E_F}{k_B T} &= \frac{E_c + E_v}{k_B T} + \ln N_v - \ln N_c \\ \Rightarrow E_F &= \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln \frac{N_v}{N_c} \end{aligned}$$

- If we substitute the expressions for N_c and N_v , in the ratio of N_c over N_v all the terms cancel out except the effective masses of electron and hole.

Fermi level of intrinsic semiconductor

- The expressions for effective density of states are

$$N_c = 2 \left(\frac{2\pi m_n^* k_B T}{h^2} \right)^{3/2}, \quad N_v = 2 \left(\frac{2\pi m_p^* k_B T}{h^2} \right)^{3/2}$$

- Therefore, the Fermi level in terms of effective masses of charge carriers is given by

$$\begin{aligned} E_F &= \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln \left(\frac{m_p^*}{m_n^*} \right)^{3/2} \\ &= E_{\text{midgap}} + \frac{3k_B T}{4} \ln \frac{m_p^*}{m_n^*} \end{aligned}$$

where E_{midgap} is defined as the midgap energy level that is in the midpoint of conduction band minimum and valence band maximum and is given by

$$E_{\text{midgap}} := \frac{E_c + E_v}{2}$$

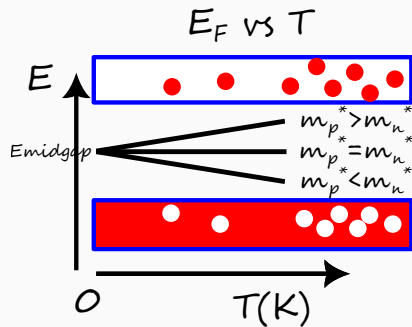
$$\therefore E_F = E_{\text{midgap}} + \frac{3k_B T}{4} \ln \frac{m_p^*}{m_n^*}$$

E_F vs T of intrinsic semiconductor

- Since the Fermi energy level varies with temperature, let us consider two cases – $T = 0\text{ K}$ and $T \neq 0\text{ K}$.
- Also, the Fermi energy level varies with effective masses of electron and hole. So let us consider three cases – $m_p = m_n$, $m_p > m_n$, and $m_p < m_n$.

	$T = 0\text{ K}$	$T \neq 0\text{ K}$
$m_p = m_n$	E_{midgap}	E_{midgap}
$m_p > m_n$	E_{midgap}	$E_{\text{midgap}} + \delta$
$m_p < m_n$	E_{midgap}	$E_{\text{midgap}} - \delta$

Table 1: Here δ is a positive quantity that linearly increases with temperature.



Key Insight

With temperature, E_F deviates from E_{midgap} for semiconductors with differing electron and hole effective masses.

Conductivity of intrinsic semiconductor

- Conduction in a semiconductor is through two channels – electron and hole. Therefore

$$\sigma_i = \sigma_e + \sigma_h$$

- For each channel, the conductivity is given by the product of charge carrier concentration and mobility. Therefore

$$\begin{aligned}\sigma_i &= n_0 e \mu_e + p_0 e \mu_h \\ &= n_i e (\mu_e + \mu_h)\end{aligned}$$

$$\boxed{\therefore \sigma_i = n_i e (\mu_e + \mu_h)}$$

- If we express the intrinsic carrier concentration in terms of material properties, then

$$\begin{aligned}\sigma_i &= \sqrt{N_c N_v} e (\mu_e + \mu_h) \exp\left(-\frac{E_g}{2k_B T}\right) \\ &= \sigma_0 \exp\left(-\frac{E_g}{2k_B T}\right)\end{aligned}$$

where σ_0 is a constant in terms of material properties.

Some useful information for calculations

	Si	GaAs	Ge
μ_n ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	1350	8500	3900
μ_p ($\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$)	480	400	1900

Table 2: Mobility of carriers in Si, GaAs, and Ge at 300 K. Taken from Section 5.1 of Neamen

	Si	GaAs	Ge
n_i (cm^{-3})	1.5×10^{10}	1.8×10^6	2.4×10^{13}

Table 3: Intrinsic carrier concentration in Si, GaAs, and Ge at 300 K.

Some useful information for calculations (continued...)

	Si	GaAs	Ge
$N_c \text{ (cm}^{-3}\text{)}$	2.8×10^{19}	4.7×10^{17}	1.04×10^{19}
$N_v \text{ (cm}^{-3}\text{)}$	1.04×10^{19}	7.0×10^{18}	6.0×10^{18}
m_n^*/m_0	1.08	0.067	0.55
m_p^*/m_0	0.56	0.48	0.37

Table 4: Effective density of states and effective mass of charge carriers in Si, GaAs, and Ge . Taken from Section 4.1.3 of Neamen.

	Si	GaAs	Ge
$E_g \text{ (eV)}$	1.12	1.42	0.66

Table 5: Energy band gap of Si, GaAs, and Ge at 300 K. Taken from Appendix. B of Neamen.

Problems on Fermi level of intrinsic semiconductor

Problem

A quantum state in the conduction band is $k_B T$ above conduction band minimum E_c of an intrinsic semiconductor. The Fermi level is 0.25 eV below E_c . Find the probability that the quantum state is occupied at 300 K.

Problem

For the same material in above problem, if the effective density of states in conduction band N_c is $2.8 \times 10^{19} \text{ cm}^{-3}$ at 300 K, then find the electron concentration n_0 .

Problem

Calculate the position of the intrinsic Fermi level with respect to the center of the bandgap in Si at 300 K.

Problem

Find the resistivity of intrinsic Ge at 300 K.

[Hint: Check useful info slide]

Problems on Fermi level of intrinsic semiconductor (continued...)

Problem

The resistivity of intrinsic Si is $2.3 \times 10^3 \Omega \text{ m}$ at 300 K. Calculate its resistivity at 100 °C.

[Hint: $E_g^{Si} = 1.12 \text{ eV}$]

Tough problems

Problem

Calculate the thermal equilibrium electron concentration in Si at 400 K. N_c at 300 K is $2.8 \times 10^{19} \text{ cm}^{-3}$.

[Hint: Solve the below problem.]

Problem

Calculate the ratio of conduction electrons at 400 K to conduction electrons at 300 K.
[Hint: From the explicit expression for N_c , observe the form of N_c vs T .]

Test Your Understanding

1. Discuss the variation of Fermi level position with temperature.
2. Discuss the variation of conductivity with temperature.

- band structure
- zone folding
- hole
- conduction band effective density of states N_c
- conduction band effective density of states N_v
- intrinsic carrier concentration n_i
- midgap energy level E_{midgap}

M2U2 Plan

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pn junction at zero bias

Definition

Zero bias condition is considered when external battery is not connected across p-region and n-region.

- The word voltage bias is used in the context of electricity to refer to external force.
- At zero voltage bias, there is **no application of external force** on the charge carriers.
- However, there are two internal forces at play –
 1. Diffusion forces due to gradients in concentration of charge carriers.
 2. Drift forces due to internal electric fields.
- The interplay of these internal forces leads to net zero force on charge carriers at thermal equilibrium.

Key Insight

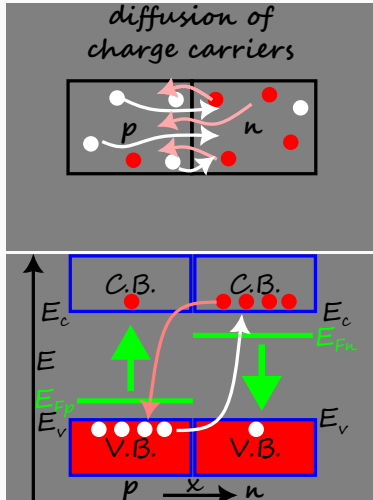
At zero bias, there is interplay of internal drift and diffusion forces.



pn junction – Diffusion

Definition

Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration.

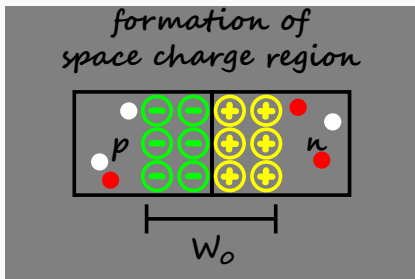


- The difference in Fermi level acts as a “driving force” so that electrons from n-region **diffuse** into the p-region.
- By similar analysis, holes from p-region diffuse into the n-region.
- Diffusion of electrons from n-region lowers its Fermi level E_{Fn} .
- Diffusion of holes from p-region raises its Fermi level E_{Fp} .

pn junction at zero bias – Space-charge region

Definition

- **Space charge region**, also called the **depletion region**, is the region created as a result of diffusion of free charge carriers across the metallurgical junction.
- The width of the depletion region is called depletion region width W_0 .



- Due to the diffusion of mobile charge carriers, their **ionic counterparts** are left without partners.
- Space charge region is also called the **depletion region** since the region is depleted ^a of the free charge carriers.

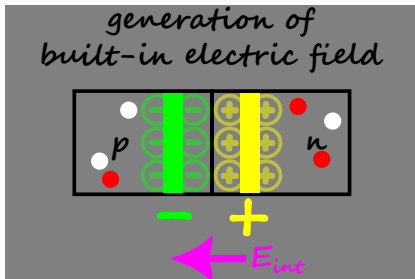
^adeplete means decrease or remove.

- The depletion region consists of positively charged donor ions on the n-side of the junction and negatively charged acceptor ions on the p-side of the junction.

pn junction at zero bias – Built-in electric field

Definition

Built-in electrical field, also called internal electric field E_{int} is the electric field generated due to the oppositely charged regions on either side of the metallurgical junction.

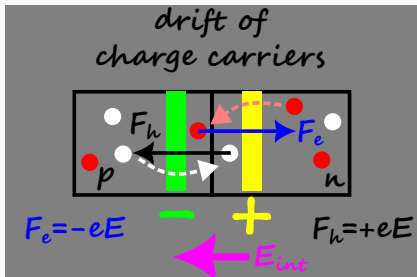


- Within the depletion region, the oppositely charged ions on either side of junction can be considered as plates of a capacitor.
 - Since the plates are charged, a **built-in electric field** is generated.
 - The built-in electric field is directed from **n-region to p-region**.
- E_{int} prevents further build up of space charge due to diffusion as it opposes the diffusion of electrons from n-region into p-region and vice versa.

pn junction at zero bias – Drift

Definition

Drift is the net movement of charged particles due to the application of electric field.



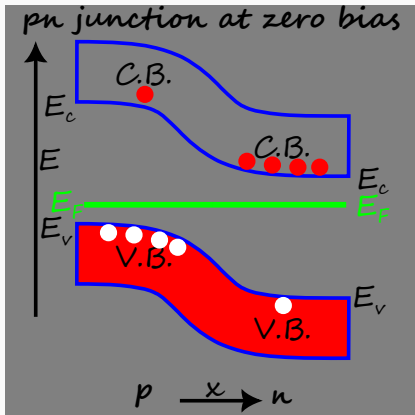
- Consider a diffused electron that is present in the p side of the depletion region.
- Since the built-in electric field is directed from n-region to p-region, the electric force is directed from p-region to n-region.
- Electron drift backs to the n-region.

- The diffusion force is counterbalanced by the drift force.
- Similar analysis applies to the diffused hole.

Problem

Prove the counterbalancing effect of drift and diffusion for holes.

pn junction at zero bias – Thermal equilibrium



Definition

Thermal equilibrium occurs when the net force on the system is zero.

- Since the difference in Fermi level is a “driving” force, when the system reaches thermal equilibrium, the net force is zero.
- This implies the Fermi level is a constant throughout the system at thermal equilibrium.

- Since the external force is zero at zero bias, the pn junction is at thermal equilibrium.

Key Insight

At thermal equilibrium, E_F is a constant.



pn junction at non-zero bias

pn junction – at non-zero bias

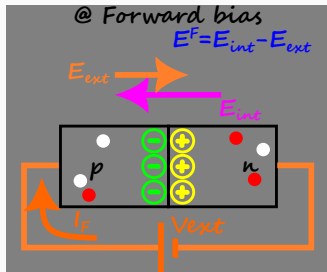
- At non-zero voltage bias, there is **an application of external force** on the charge carriers. As a result, the pn junction cannot attain thermal equilibrium.
- Since there is no thermal equilibrium, the Fermi energy is not a constant across the pn junction.
- Since the direction of built-in electric field is from n-region to p-region, the behaviour of p-n junction is different when the external field E_{ext} is parallel or anti-parallel to the built-in electric field E_{int} .
 1. $E_{\text{ext}} \uparrow \downarrow E_{\text{int}}$ corresponds to forward biasing.
 2. $E_{\text{ext}} \uparrow \uparrow E_{\text{int}}$ corresponds to reverse biasing.

Key Insight

At non zero bias, the Fermi energy is not a constant across the pn junction.



pn junction – at forward bias



Forward

$$V_F > 0$$

$$E^F < E_{\text{int}}$$

$$W_F < W_0$$

$$V_F < V_{\text{int}}$$

$$E_{F_n} - E_{F_p} > 0$$

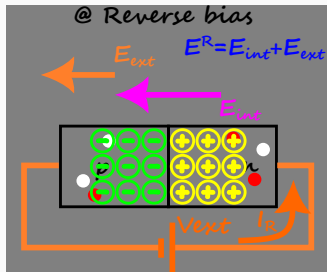
$$I_F > 0$$

Definition

Forward bias condition is considered when the positive electrode of battery with potential V_F is connected to p-region and negative electrode is connected to n-region.

- The external bias $V_{\text{ext}} = V_F$ is from p-region to n-region.
- The external field E_{ext} is from p-region to n-region. The internal field E_{int} is **anti-parallel** to E_{ext} . The net electric field E^F is lower than the case of zero bias.
- So less space charge is created in depletion region. So we have smaller space charge width W_F .

pn junction – at reverse bias



Reverse

$$V_R < 0$$

$$E^R > E_{\text{int}}$$

$$V_R > V_{\text{int}}$$

$$E_{Fn} - E_{Fp} < 0$$

$$W_R > W_0$$

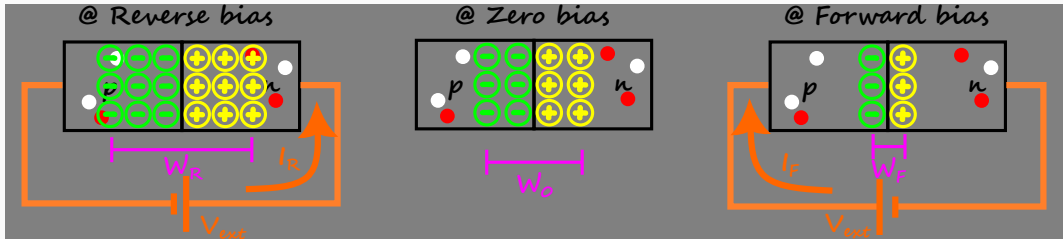
$$I_R < 0$$

Definition

Reverse bias condition is considered when the positive electrode of battery is connected to n-region and negative electrode is connected to p-region.

- The external bias $V_{\text{ext}} = V_R$ is from n-region to p-region.
- The external field is from n-region to p-region.
Under reverse bias condition, the internal field is **parallel** to the external field. The net electric field E^R is higher than the case of zero bias.
- So more space charge is created in depletion region. So we have larger space charge width W_R .

Comparison of depletion region widths at different biases

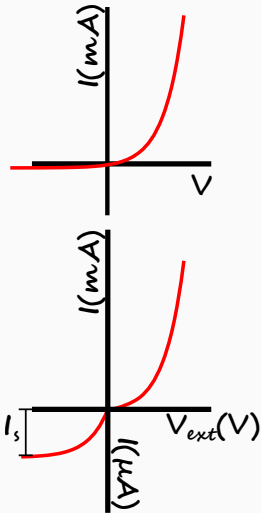


$$W_R > W_0 > W_F$$

Large forward current vs tiny reverse current

- Under biasing conditions, the electrodes of the battery change the Fermi levels of the p and n regions.
- Since the negative electrode supplies electrons, it raises the Fermi level of the connected region. Similarly, positive electrode lowers the Fermi level.
- Under forward biasing, the energy of electrons at n-region is raised. The flow of electrons to lower energy p-region is possible. Similarly, the energy of holes at p-region is lowered. The flow of holes to higher energy n-region is possible. This leads to **large forward current**.
- Under reverse biasing, the energy of electrons at n-region is lowered. The flow of electrons to higher energy p-region is difficult. Similarly, the energy of holes at p-region is raised. The flow of holes to lower energy p-region is difficult.
- However, the reverse bias is favourable for minority electrons in p region and minority holes in n-region. Since the number of these carriers is very few, we see a **tiny reverse current**.

pn junction – $I - V$ characteristics



Definition

Rectifying action of pn junction is its property to allow **uni-directional** conduction.

- The diode equation can be shown to be

$$I = I_s \left[\exp \left(\frac{eV}{k_B T} \right) - 1 \right]$$

where I_s is the reverse saturation current.

- The equation and the $I - V$ capture the non-linear nature of p-n junction.
- In the forward bias condition, it acts as a “closed switch” allowing large current to flow through it.
- In the reverse bias condition, it acts as an “open switch” allowing insignificantly low current to flow through it.
- This unidirectional conduction is called rectifying action.

Conceptual Questions

Problem

What causes majority carriers to flow at the moment when p-region and n-region are brought together? Why does this flow not continue until all carriers have recombined?

[Hint for second part: Convince yourself that the built-in electric field opposes the diffusion “forces” for both the electrons and holes.]

Tough problems

Problem

Prove that the analysis of pn junction using potential energy barrier maps with the analysis using Fermi energy level. In other words, prove that $E_{Fn} - E_{Fp} = eV_{\text{ext}}$.

Test Your Understanding

1. What causes majority carriers to flow at the moment when p-region and n-region are brought together? Why does this flow not continue until all carriers have recombined? [Hint for second part: Convince yourself that the built-in electric field opposes the diffusion “forces” for both the electrons and holes.]
2. Explain the formation of **potential barrier** across the pn junction region.
3. Explain the formation of **depletion region** in the pn junction region. What is the typical value of depletion region width for commonly used pn junctions?
4. Draw band diagrams for symmetrically doped pn junction at
 - zero bias
 - forward bias
 - reverse bias
5. What is reverse saturation current in a diode?
6. Why is reverse saturation current independent of reverse bias?
7. What is reverse breakdown?

Technical terms

- metallurgical junction
- diffusion
- built-in electric field
- thermal equilibrium
- built-in potential barrier
- depletion region
- space charge region
- space charge width
- reverse saturation current
- reverse breakdown

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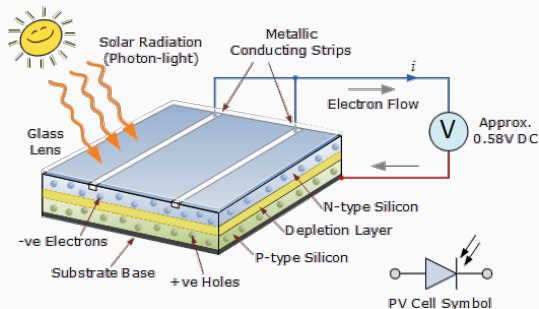
- Electronics is the study of controlling electrical signals.
- Any electrical signal is characterized by its current and voltage properties.
- Electronics is broadly classified into three areas –
 - **Microelectronics** – here we study switching electrical signals, which is useful to represent binary logic. This is necessary for computation. This is also called **digital electronics**. Since power is not a concern, signals have currents of the order of microamperes (hence the name microelectronics) and voltages of the order of few volts
 - **Power electronics** – here we study amplifying electrical signals to transfer energy. This is necessary for energy requirements of society as it is well known that electrical energy is efficient way of energy transfer. Signals have currents of the order of hundreds of amperes and voltages of the order of thousands of volts. This is also called **analog electronics**.
 - Opto-electronics ⁶ – here we harness the interaction between electron and photon. The advantages of light like its high velocity, free source of energy (Sun light), are combined with the advantages of electron like its easy manipulation to develop devices of specific requirements.

⁶Opto-electronics is a combination of optics and electronics

Solar cell – construction

Definition

- Solar cell, also technically called the **photovoltaic cell** is an opto-electronic device based on pn junction that can generate electrical power when illuminated.
- A collection of cells in series and parallel combination is called a solar panel



Sketch credit: Alternative Energy Tutorials

Sketch of solar panel. Sketch taken from Alternative Energy Tutorials

- The main aim of solar cell operation is to convert solar energy into electrical energy.
- Electrodes are deposited on both sides of junction. The top metal electrode is in the form of metal grid with fingers, which permits sunlight to pass through it.

Solar cell – working principle

Definition

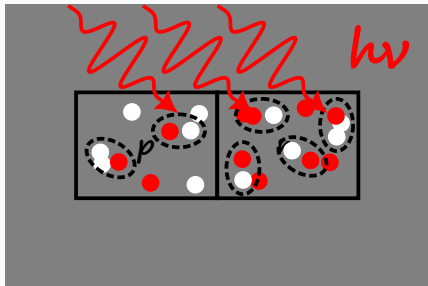
The working principle of solar cell involves three processes–

1. Generation of **excess** electron-hole pairs upon the illumination of light, and
2. Separation of electrons from holes **before** radiative recombination due to the built-in electric field.
3. Generation of voltage due to **accumulation** of excess separated charge carriers at the ends of the pn junction
 - The solar cell pn junction is operated without bias i.e. zero bias.
 - Upon illumination with photons of suitable energy, the photon are absorbed through the surface area. The covalent bonds of the lattice are broken releasing the free electrons and holes into the conduction band and valence band respectively.

Problem

A material has bandgap of 5 eV. Can it be used as solar cell material? Why or why not?

Working principle – 1. Excess electron-hole pairs



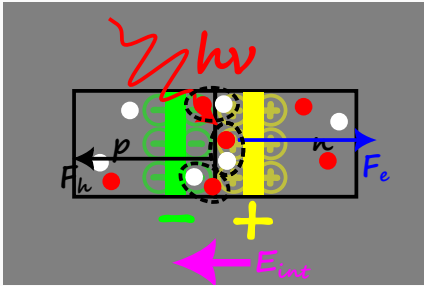
- Upon illumination with photons with energy greater than the band gap, optical absorption takes place.
- This results in the generation of excess electron-hole pairs throughout the device.

Key Insight

Optical absorption generates excess electron-hole pairs.



Working principle – 2b. Illumination at depletion region



- At the depletion region also, the excess charge carriers are generated.
- However, before they undergo radiative recombination, the built-in electric field sweeps them out of the depletion.

Key Insight

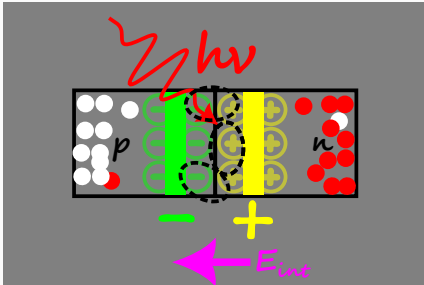
At the depletion region, separation of excess charge carriers prevents radiative recombination.



Working principle – 3. Photo-voltage

Definition

Photovoltage, also called photo e.m.f.⁷ is the voltage appearing across the device due to the separation of excess charge carriers.



- The swept excess electrons and holes **accumulate** at the n region and p region edges respectively
- The accumulation of positive charge at the p-edge and negative charge at the n-edge leads to the generation of a voltage V_L .

- Since the source of voltage is illumination of photons, it is called **photovoltage**.

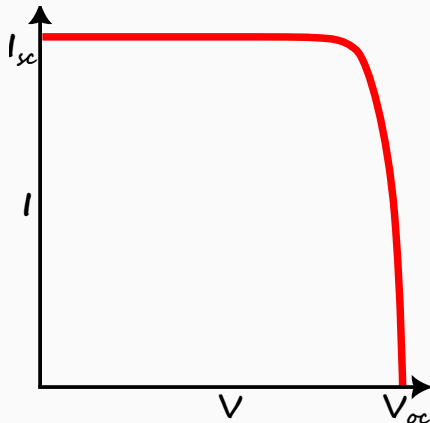
Key Insight

The photovoltage forward biases the pn junction.



⁷e.m.f. stands for electro motive force

Solar cell – $I - V$ characteristics



- From Kirchoff voltage law, the photo-voltage is given by

$$V_L = I_L R_L$$

- From diode $I - V$ characteristics, the forward current I_F is given by

$$I_F = I_s \left[\exp \left(\frac{eV_L}{k_B T} \right) - 1 \right]$$

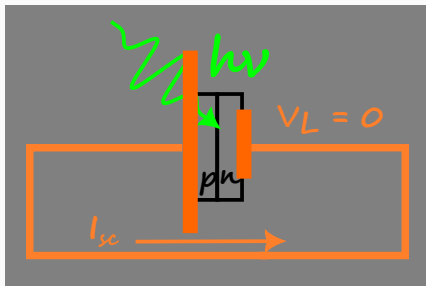
- Therefore the solar cell $I - V$ characteristics is given by

$$\begin{aligned} I_{\text{net}} &= I_L - I_F \\ &= I_L - I_s \left[\exp \left(\frac{eV_L}{k_B T} \right) - 1 \right] \end{aligned}$$

Solar cell – short circuit current

Definition

Short circuit current is defined the current through solar cell when it is short circuited or zero load resistance is connected.



- When the load resistance is zero, the solar cell generates maximum current. This condition is called short circuit condition.
- At short circuit condition, $R_L = 0$, so the voltage drop across load is zero

$$V_L = I_L R_L = 0$$

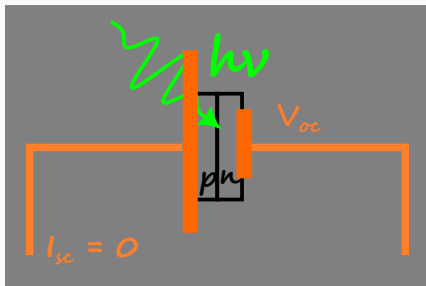
- The net current is the short circuit current I_{sc} and is given by

$$\begin{aligned} I_{sc} &= I_L - I_s (e^0 - 1) \\ &= I_L \end{aligned}$$

Solar cell – open circuit voltage

Definition

Open circuit voltage is defined the voltage across solar cell when it is open circuited or infinite load resistance is connected.



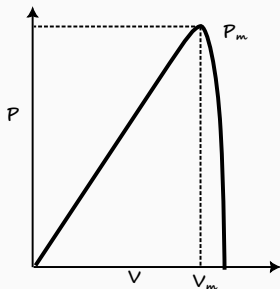
- When the load resistance is infinite, the solar cell generates maximum voltage. This condition is called open circuit condition.
- At open circuit condition, $R_L = \infty$, so the current is zero

$$I_L = \frac{V_L}{R} = 0$$

- The photovoltage is the open circuit voltage V_{oc} and is given by

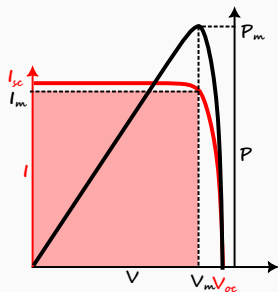
$$I_L - I_s \left[\exp \left(\frac{eV_{oc}}{k_B T} \right) - 1 \right] = 0 \Rightarrow I_L = I_s \left[\exp \left(\frac{eV_{oc}}{k_B T} \right) - 1 \right]$$
$$\exp \left(\frac{eV_{oc}}{k_B T} \right) = \frac{I_L}{I_s} + 1 \Rightarrow V_{oc} = \left(\frac{k_B T}{e} \right) \ln \left(1 + \frac{I_L}{I_s} \right)$$

Solar cell – $P - V$ characteristics



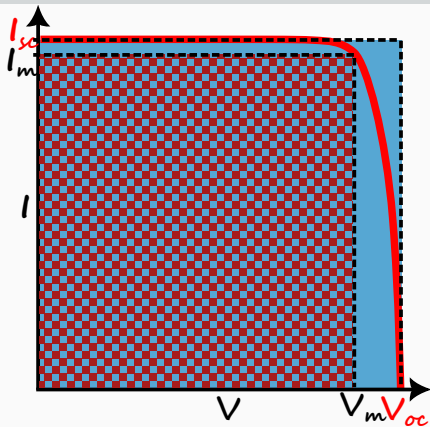
- The power generated in solar cell is completely consumed in the load resistance.
- If the photocurrent at V_m is I_m then the maximum power is

$$P_m = V_m I_m$$



- In $P - V$ curve, this is represented as (V_m, P_m) .
- The I & $P - V$ curves show the mappings of $I - V$ and $P - V$ curves. In $P - V$ curve, maximum power is represented as as red rectangle.

Output electrical power – Fill factor



- From the $I - V$ characteristics of solar cell, it is evident that the maximum current generated cannot exceed the short circuit current I_{sc} and the maximum voltage generated cannot exceed the open circuit voltage V_{oc} .
- The theoretically possible maximum power that can be generated should be less than the product of I_{sc} and V_{oc} . In the $I - V$ curve, this is represented as blue rectangle.

- The ratio of experimentally possible maximum power to theoretically possible maximum power is called Fill Factor and is given by

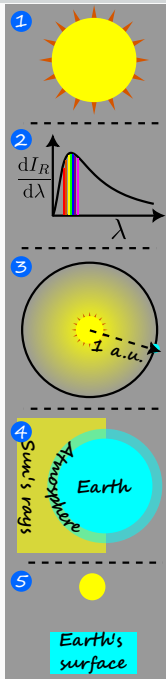
$$FF = \frac{V_m I_m}{V_{oc} I_{sc}}$$

Key Insight

Fill factor is a material property.



Input optical power – Solar irradiance



Definition

Solar irradiance is the solar power incident on the Earth's surface per unit area.

1. Sun as blackbody – Since sun is the major source of light energy, the input power is determined by the properties of sun.
2. Planck's blackbody radiation law – Sun is emitting photons of various wavelengths per unit time. Each photon with a given wavelength has an energy given by the Planck's law.
3. Sun -Earth separation – However, we are only receiving a fraction of the Sun's power.
4. Solar irradiance outside atmosphere – It is 1360 W m^{-2} .
5. Solar irradiance at Earth's surface – It is nearly 1000 W m^{-2} at noon due to losses in atmosphere.

Practical considerations – Solar cell efficiency

Definition

Conversion efficiency is defined as ratio of output electrical power to input optical power.

- To harness solar energy, the following practical limitations need to be considered.
- All the power cannot be harnessed as the optical absorption by a solar cell has a cut off wavelength due to the energy band gap.
- The photons with energy greater than energy band gap also generate electron-hole pairs. However, the difference of photon energy and band gap is dissipated as heat.
- The sum total of all these considerations is that a fraction of incident optical power is converted into electrical power and is given by the conversion efficiency η as

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100$$

Solar cell – applications

Chandrayaan 2 mission at glance [Click for video]



Image credits: ISRO

Problem

From the data in above figure, estimate the conversion efficiency of material used in Pragyan rover.

Test Your Understanding

1. Explain the working of solar cell.
2. Draw and explain the $I - V$ characteristics of a solar cell.
3. A material has bandgap of 5 eV. Can it be used as solar cell material? Why or why not?
4. An alien ship crashlanded onto Earth's surface. From the debris, the solar panel used for power generation is found to have a band gap of 5 eV. Estimate the temperature of the alien Sun.

- optical absorption
- generation of excess charge carriers
- conversion efficiency
- fill factor
- open-circuit voltage
- short-circuit current

M2U2 Plan

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p-n junction diode – forward and reverse conditions

Solar cell

Photo-diode

Direct and indirect band gap semiconductors

Light Emitting Diode

Laser

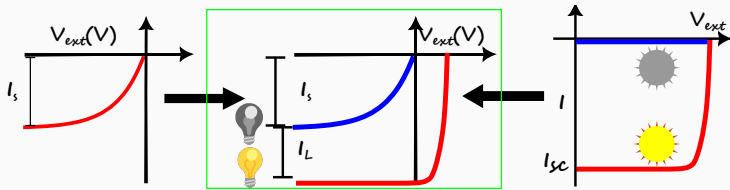
Optical fiber

Definition

A photo-diode is an opto-electronic device that is used to sense the presence of light.

- A photo-diode is a pn junction diode that operates under **reverse bias**.
- Due to the reverse bias, the depletion region width increase. This leads to decrease in junction capacitance.

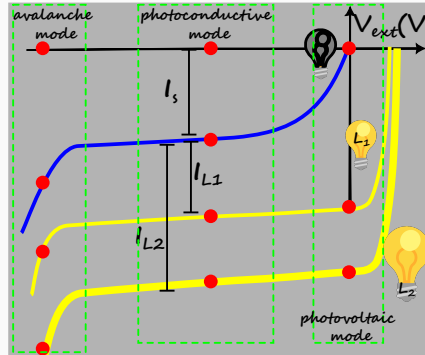
Photo-diode – $I - V$ characteristics



- The $I - V$ characteristics of photodiode can be considered as $I - V$ characteristics of solar cell under forward bias and $I - V$ characteristics of pn junction under reverse bias.
- The reverse bias generates reverse current that saturates to reverse saturation current I_s at high bias.
- In addition to the reverse current, if the photons are incident on the diode there is an additional photocurrent I_L .
- Thus the external bias is negative and the net current is negative and the $I - V$ characteristics is given by

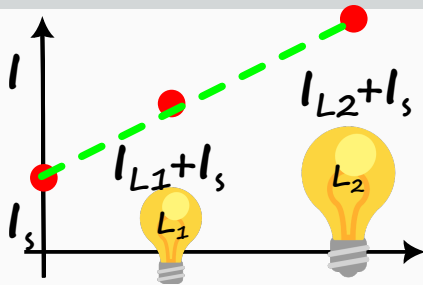
$$I = I_s \left[\exp \left(\frac{eV}{k_B T} \right) - 1 \right] - I_L$$

Photo-diode – Modes of operation



- The photo diode has three modes of operation
 1. photoconductive mode – reverse bias
 2. zero bias mode – reverse bias voltage is set to zero
 3. avalanche mode – **very** high reverse bias
- In every mode of operation, there are two states of operation
 1. Photons are not incident 💡
 2. Photons are incident 💡

Photoconductive mode



- Photo diode is under reverse bias.
- At zero light intensity, there is a non zero current due to the reverse saturation current. This is called **dark current**.
- Response of photodiode is linear. In this mode, photodiode is a linear sensor.

- Due to reverse bias, the depletion region width increases. Since the space charge region acts as a parallel plate capacitor, the junction capacitance C_{junction} is given by

$$C_{\text{junction}} = \frac{\epsilon A}{W}$$

where A is the cross sectional area of the depletion region and W is the depletion region width.

- Due to reduction in junction capacitance, photodiode is used for high frequency opto-electronic converters.

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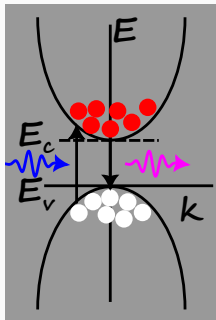
Laser

Optical fiber

Direct band gap semiconductor

Definition

A direct band gap semiconductor is defined as a semiconductor with the top of valence band and bottom of conduction band at the **same** value of wave vector.



- Consider electron moves from E_c to E_v .
- In the radiative recombination process,
 - Energy of electron is transferred to energy of photon.
 - Initial momentum of electron is equal to final momentum.
 - Therefore, both the energy and momentum of the (electron + hole + photon) system is **conserved**.
 - **three** particles are involved



Key Insight

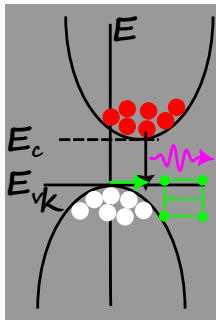
Radiative type of recombination is the dominant mechanism in a direct bandgap semiconductor.



Indirect band gap semiconductor

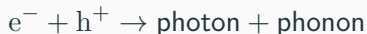
Definition

An indirect band gap semiconductor is defined as a semiconductor with the top of valence band and bottom of conduction band at the **different** values of wave vector.



- Consider electron moves from E_c to E_v .
- In the radiative recombination process,
 - Energy of electron is transferred to energy of photon.
 - Initial momentum of electron is not equal to final momentum.
 - Therefore, energy of the (electron + hole + photon) system is **conserved** but the momentum is not conserved.

- To conserve momentum, a **phonon assisted radiative recombination** occurs where difference in momentum is supplied by phonon and **four** particles are involved



Direct vs indirect bandgap semiconductor

SNo.	Property	Direct	Indirect
1	k at E_C vs k at E_V	equal	unequal
2	Recombination mechanism	Radiative	Non-radiative
3	Particle emitted	Photon	Phonon
4	Periodic table	Compound s.c. like GaAs, InP	Elemental s.c. like Si, Ge
5	Applications	LED, laser diode	transistors, diodes

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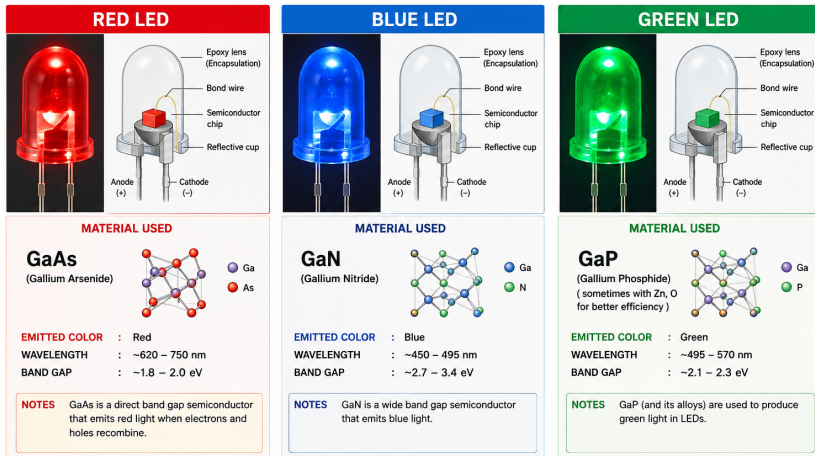
Definition

Light Emitting Diode (LED) is the opto-electronic device based on radiative recombination of charge carriers.

- LED is based on pn junction diode under forward bias.
- Under forward bias, the electrons and holes crossing the depletion region undergo radiative recombination leading to generation of light.
- From the perspective of electroluminescence, it is useful to use a semiconductor with a direct band gap semiconductor.

LED – construction

LED COLORS AND SEMICONDUCTOR MATERIALS USED

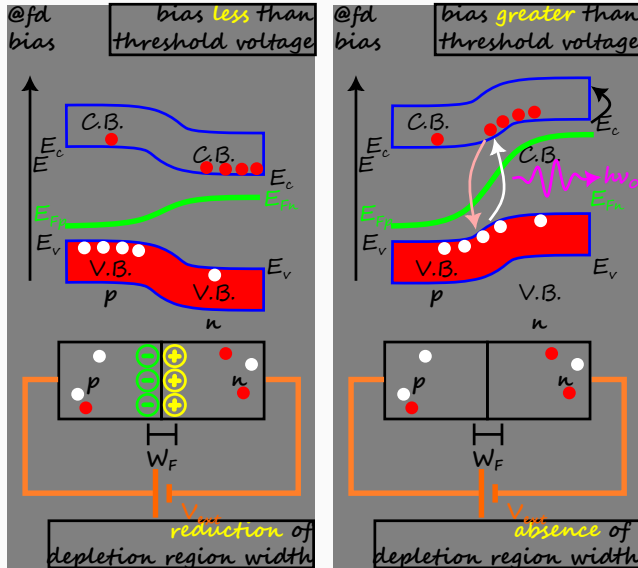


SUMMARY | LED color depends on the semiconductor material and its band gap.

Higher band gap → shorter wavelength (blue) | Lower band gap → longer wavelength (red)

- New materials AlGaInP for Red, $\text{In}_x\text{Ga}_{1-x}\text{N}$ for Green, and $\text{In}_x\text{Ga}_{1-x}\text{N}$ for Blue.

LED – working principle



- LED works on the principle of **electroluminescence**.
- Below threshold voltage, potential energy barrier is reduced, and depletion region width is reduced.
- Above threshold voltage, potential energy barrier is removed, and depletion region width is removed. The electrons and holes move towards the junction, and **recombine radiatively** to emit photon.

LED – optical emission

- From the Planck's quantum hypothesis, the wavelength can be related to the energy of photon by

$$\begin{aligned} E &= h\nu, \\ &= \frac{hc}{\lambda}, \\ E \text{ (in eV)} &= \frac{1.24}{\lambda \text{ (in } \mu\text{m)}} \end{aligned} \quad (\text{A})$$

Problem

Prove Equation. A.

Problem

Find the energy of red photon with wavelength 700 nm.

Problem

Find the energy of blue photon with wavelength 400 nm.

- Night lighting
- Traffic signals
- Optical fibre communications
- Medical imaging

Summary of opto-electronic devices

- From the perspective of bias

Reverse bias	Zero bias	Forward bias
--------------	-----------	--------------

Photo-diode	Solar cell	LED
-------------	------------	-----

- From the perspective of depletion region width

Larger width	Nominal width	Smaller width
--------------	---------------	---------------

Low current	Nominal current	Large current
-------------	-----------------	---------------

- From the perspective of power consumption

Low Power	Nominal Power	High Power
-----------	---------------	------------

Consumption	Generation	Consumption
-------------	------------	-------------

- From the perspective of capacitance

Low Capacitance	Nominal Capacitance	High Capacitance
-----------------	---------------------	------------------

High frequency	—	—
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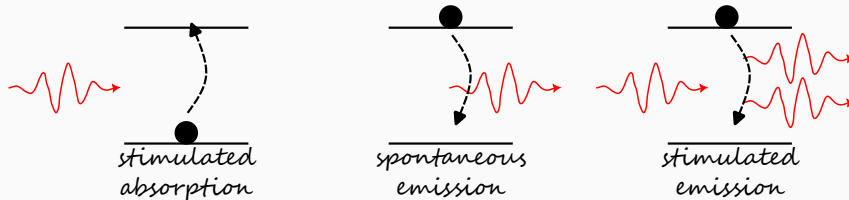
Light Emitting Diode

Laser

Optical fiber

Laser – Three Quantum Processes

- L.A.S.E.R is acronym for Light Amplification by Stimulated Emission of Radiation.
- Einstein predicted the possibility of lasing in 1916 by introducing concept of stimulated emission.



- Light interacts with matter by three quantum processes
 - Stimulated absorption
 - Spontaneous emission
 - Stimulated emission

Laser – Three Quantum Processes (contd...)

Definition

Stimulated absorption occurs when electron transitions from lower energy level to higher energy level upon the absorption of photon.



Definition

Spontaneous emission occurs when electron transitions from higher energy level to lower energy level by the emission of photon.



Definition

Stimulated emission occurs when electron transitions from higher energy level to lower energy level in the presence of a photon by the emission of additional photon.



Laser – Conditions and requirements for lasing action

From the Einstein's theory ⁸ of light-matter interaction, the conditions for lasing action are –

- high photon density,
- large lifetime of excited levels, and
- high population of excited levels to achieve lasing action.

To satisfy these three conditions for lasing action, the requirement of lasing action are –

1. Population inversion
2. Meta-stable energy level
3. Radiation confinement

⁸Refer to Chapter 44 Avadhanulu for a discussion on this theory.

Thermal equilibrium and normal state

Definition

Maxwell-Boltzmann statistics says the probability of occupying higher energy level is lower than the probability of occupying lower energy level. The probability of occupying energy level E is proportional to

$$P_{MB} \propto \exp\left(-\frac{E}{k_B T}\right)$$

Definition

Normal state is defined as the state under thermal equilibrium.

- Under the conditions of thermal equilibrium, the occupancy of energy levels is governed by Maxwell Boltzmann statistics.



Problem

Find the ratio of populations of two states in a He–Ne laser that produces light of wavelength 6328 Å at 27 °C.



normal state 220 / 252

To satisfy the above three requirements of lasing action, we need three components to any laser –

1. Active medium
2. Optical pump
3. Optical resonant cavity

Laser – characteristics

Laser light 	Ordinary light 
Unidirectional	Non-directional
Coherent	Incoherent
Monochromatic	Non-monochromatic
High intensity	Low intensity
Low pulse width	High pulse width

- Coherent means photons in the beam have same phase
- Monochromatic beam consists of photons of single color/wavelength.

Diode laser – Construction

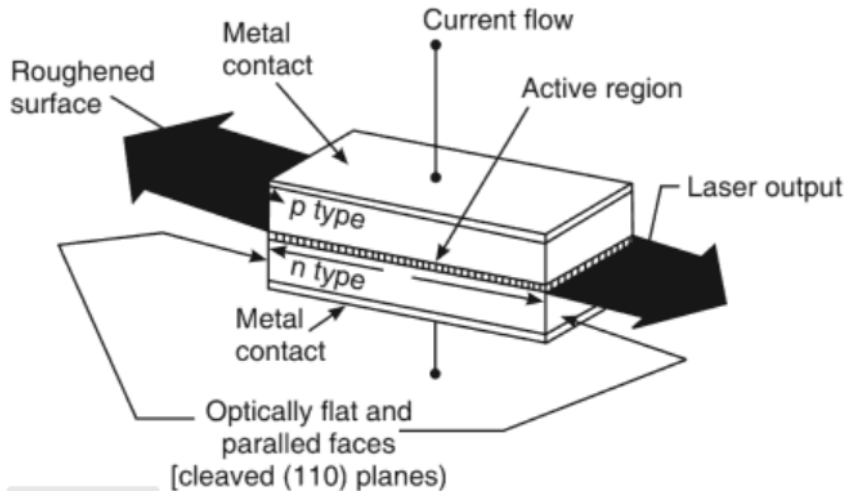
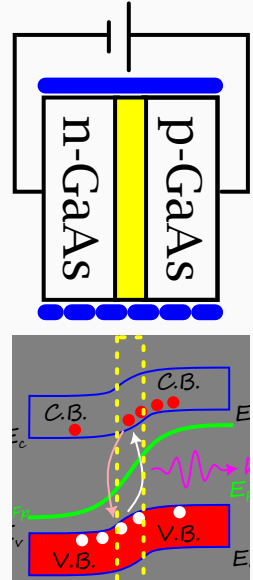


Fig. 44.24:

Schematic of homojunction diode laser

Diode laser – working principle

- In a diode laser
 - Depletion region is the active medium,
 - Forward biasing is the pump, and
 - Polished faces of the GaAs crystal are the optical resonator.
- Since the forward biasing populates the electrons in the n region and holes in the p region, when they meet at the depletion region, the active medium is already in the lasing state.
- Therefore, there is no need for a meta-stable state.



Diode laser – applications

- Industrial precision cutting
- Lasik surgery
- LiDAR (Light Imaging, Detection and Ranging) technology for distance measurement
- Optical fibre communication technology

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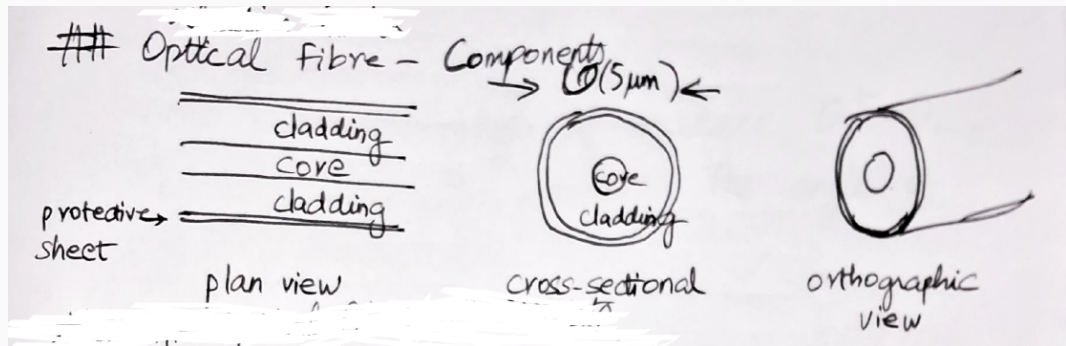
Laser

Optical fiber

Optical fiber – light pipe

- Optical fiber is an opto-electronic device that is the backbone of modern communication systems.
- Useful for long distance communication.
- Harnesses the speed of light and property of light propagation across media with different refractive indices.

Optical fiber – construction



Optical fiber – working principle

- The speed of light varies in different media.
- It is highest in vacuum and is given by $v = c$.
- In other medium, it is quantified by the refractive index n , so that the velocity of light in the medium is

$$v = \frac{c}{n}.$$

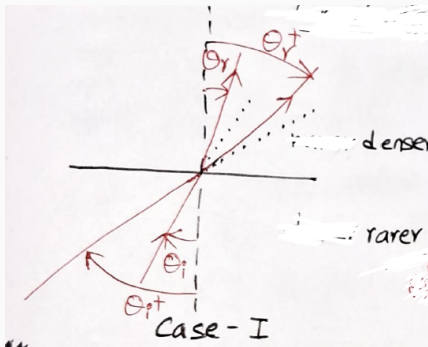
- At the interface of media, speed of light changes leading to two cases.
 - Case I: Light propagates from rarer to denser medium.
 - Case II: Light propagates from denser to rarer medium.

medium	n
air	1
water	1.33
glass	1.5

Table 6: Refractive indices of commonly available media.

Case I: Rarer to denser medium

Consider light is incident at an angle of incidence θ_i with respect to normal.

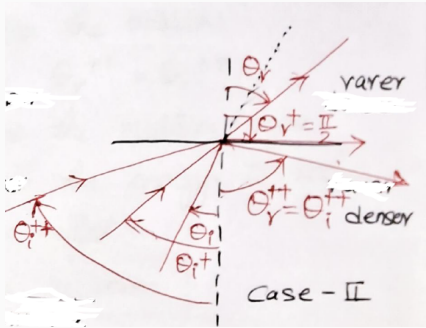


- Since light is faster in rarer medium, and slower in denser medium, it bends towards the normal. Therefore, it refracts.
- Angle of refraction is less than angle of incidence,

$$\theta_r < \theta_i.$$

- Similarly, for larger angle of incidence θ_i^+ , light refracts with angle of refraction $\theta_r^+ < \theta_i^+$.

Case II: Denser to rarer medium



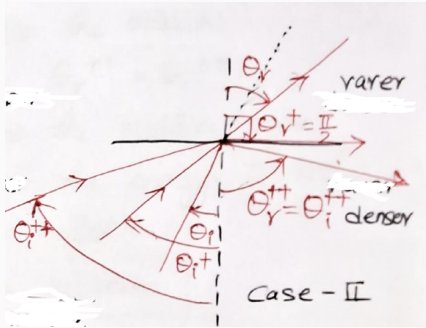
- Since light is slower in denser medium and faster in rarer medium, it bends away from the normal.
- Case (i) Angle of refraction is more than angle of incidence,

$$\theta_r > \theta_i.$$

- At a critical angle of incidence $\theta_i^+ = \theta_c$, the light grazes the interface. The angle of refraction is

$$\theta_r^+ = \frac{\pi}{2}$$

Total internal reflection

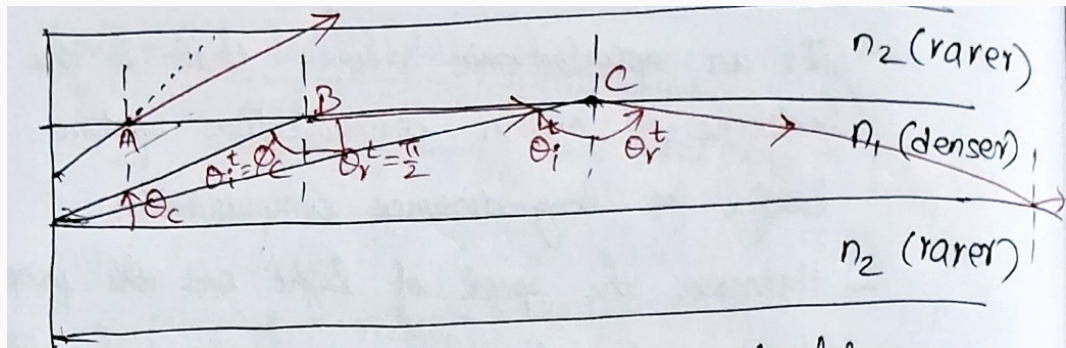


- Case (iii) For angle of incidence $\theta_i^{++} > \theta_c$, the light reflects back into the medium. The angle of reflection

$$\theta_r^{++} = \theta_i^{++}.$$

- Since the light enters the medium again with reflection and all the light energy is within the medium, this condition is called **Total Internal Reflection (TIR)**.
- TIR is required for light containment within the optical fiber. This possible if refractive index of core is greater than refractive index of cladding.

Optical fiber – conditions

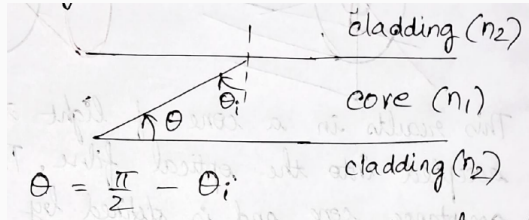


- Light is at one end of the optical fiber.
- Not all light can be contained within the light pipe (optical fiber).

Critical angle of propagation

Definition

Angle of propagation is defined as the angle between incident ray and core-cladding interface.



Acceptance angle

- Before the light enters the core medium, it has to encounter the interface between the core and the ambient ⁹ medium.
- If the refractive index of ambient medium is n_3 , then for entering the core medium, the refractive index of ambient must be lesser than refractive index of core.
- For convenience, let us consider $n_3 = 1$. Therefore, the angle of incidence at the ambient-core interface needs to be less than the so called acceptance angle so that the refracted light is within the critical angle of propagation.
- At the acceptance angle of incidence θ_0 ,

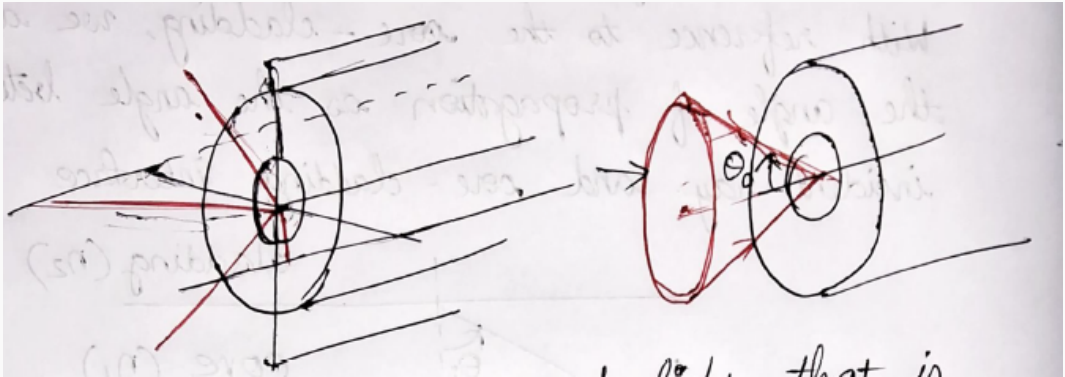
$$\frac{\sin \theta_0}{\sin \theta_0} = \frac{n_1}{n_3} = n_1$$

$$\theta_0 = \sin^{-1} \sqrt{n_1^2 - n_2^2}$$

⁹ambient means outside or commonly occurring, in our case it is usually air.

Acceptance cone

- We have limited our previous analysis to the "top" and "bottom" interface.
- Due to the cylindrical nature of optical fiber, the analysis is applicable to all planes passing through the axis.



This results in a cone of light that is accepted into the optical fiber. This is called acceptance cone and is defined by its acceptance angle.

- The industry has defined new quantities to quantify an optical fibre –
 1. numerical aperture
 2. fractional refractive index

Definition

Fractional refractive index is defined as relative difference in the refractive indices of core and cladding.

- The fractional refractive index is given by

$$\Delta = \frac{n_1 - n_2}{n_1}.$$

- In order to guide light effectively through a fiber, $\Delta \ll 1$. Typically, Δ is of the order of 0.01.

Numerical aperture

Definition

Numerical aperture is defined as the sine of the acceptance angle

$$\begin{aligned}\text{N.A.} &= \sin \theta_0, \\ &= \sin \left(\sin^{-1} \sqrt{n_1^2 - n_2^2} \right), \\ &= \sqrt{n_1^2 - n_2^2}, \\ &= \sqrt{(n_1 + n_2)(n_1 - n_2)}, \\ &= \sqrt{\left(\frac{n_1 + n_2}{2} \right) \left(\frac{n_1 - n_2}{n_1} \right)} \cdot 2n_1, \\ &\simeq \sqrt{n_1 \Delta 2n_1}, \\ &= n_1 \sqrt{2\Delta}.\end{aligned}$$

Optical fiber based communication

- Optical fiber is the backbone of modern communication system.

